

EM and ELBO

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If you have any questions, feel free to contact me(above email).

The EM algorithm maximizes a lower bound of the marginal likelihood $\Pr(\mathcal{D}; \theta)$.

It is correct that there could have a “log” before the term “marginal likelihood $\Pr(\mathcal{D}; \theta)$ ”.

But it is **incorrect** that there should be a “expected” before the term “marginal likelihood $\Pr(\mathcal{D}; \theta)$ ”.

Notice: Please classify the difference between a random variable X , and its value x . It is customary to use uppercase letters to represent random variables and lowercase letters to represent values.

$$\log p(X) = \log \frac{p(X, z)}{\frac{p(X, z)}{q(z)}} = \log \frac{p(X, z)}{q(z)} - \log p(z | X)$$

Since $\int q(z) dz = 1$, and $\log p(X)$ is a constant, so

$$\log p(X) = \log p(X) \cdot \int q(z) dz = \int q(z) \log p(X) dz$$

Thus,

$$\begin{aligned} \log p(X) &= \int q(z) \log \frac{p(X, z)}{q(z)} dz + \int q(z) \log \frac{q(z)}{p(z | X)} dz \\ &= \underbrace{\int q(z) \log \frac{p(X, z)}{q(z)} dz}_{\text{Evidence Lower Bound(ELBO)}} + \text{KL}(q(Z) \| p(Z | X)) \end{aligned}$$

In the last line, since KL divergence is non-negative, so

$$\int q(z) \log \frac{p(X, z)}{q(z)} dz$$

is a lower bound of $\log p(X)$, $\int q(z) \log \frac{p(X, z)}{q(z)} dz$ is called evidence and

$$\int q(z) \log \frac{p(X, z)}{q(z)} dz$$

is called evidence lower bound(ELBO).

We can further write the ELBO as:

$$\begin{aligned} \int q(z) \log \frac{p(X, z)}{q(z)} dz &= \int q(z) \log p(X, z) dz - \int q(z) \log q(z) dz \\ &= \int q(z) \log p(X, z) dz + \underbrace{H(q(Z))}_{\text{entropy of } q(Z)} \\ &= \mathbb{E}_{q(Z)}[\log p(X, Z)] + H(q(Z)) \end{aligned}$$

And in the M step of EM, we approximately maximize $\mathbb{E}_{q(Z)}[\log p(X, Z)]$ as maximizing the ELBO, which is the lower bound of the log marginal likelihood $\log p(X)$.